

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-0921

Malaspina 69 kV
December 26, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0921. The Interconnection Customer proposes a 64.1 MW solar photovoltaic facility interconnecting at the Malaspina 69 kV substation in the North Carolina area. The request was received on January 14, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 274 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$1,533,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Cheboygan-Mecklenburg Power Sta 138 KV	Mecklenburg Power Station-Person 345 KV L	144.4	209	5.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cheboygan-Mecklenburg Power Sta 138 KV	Rebuild line with higher-capacity conductor	\$10,508,000	15% (\$1,533,000)
	Total allocated to project		\$1,533,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.